3fish_farm.md

Problem 5 - Henry’s Fish Farm (***)

Henry Kellett keeps a fish farm. He keeps a daily record of the population of his fish. On the 1st day, he had 2 fish. On the 2nd day, the two fish had a baby fish. There were now 3 fish.

Then, the fish’s population started to grow in a strange pattern. On the 3rd day, there were 5 fish. On the 4th day, there were 8 fish.

In fact, the fishes’ population continued to grow in this strange pattern:

2 3 5 8 13 21 34 ...

You are given a day. Predict and print the number of fish on that day.

Input Format

A single integer, the required day.

Output Format

A single integer, the population of fish on the required day.

Example 1

Input:

```
1
```

Output:

```
2
```

Example 2

Input:

```
8
```

Output:

```
55
```

Note: When taking input, do not print any message. For example, `input()` or `input(“”)` would be fine, but `input(“Number: “)` or `input(“Something“)` might fail the tests.

1even_apples.md

Problem 3 - Henry’s Even Apples (*)

Henry Kellett wants to know how many of his apple boxes have an even number of apples. Output the number of apple boxes with an even number of apples.

Input Format

You are first given the number of boxes, and each following line will have the number of apples in each box.

Output Format

A single number, the number of boxes with an even number of apples.

Example 1

Input:

```
4
2
3
1
4
```

Output:

```
2
```

Explanation:

The first line represents the number of boxes - 4. Then the next four lines represents the number of apples in each box (2, 3, 1, 4). The first and last of these boxes have an even number of apples. Thus the output is 2.

Example 2

Input:

```
3
1
3
5
```

Output:

```
0
```

Note: When taking input, do not print any message. For example, `input()` or `input(“”)` would be fine, but `input(“Number: “)` or `input(“Something“)` might fail the tests.

2signal_flags.md

Problem 41 - Signal Flags (**)

Captain Kellett uses two signal flags to communicate with other ships! Each flag can be held in one of three positions: UP, MID (middle), or DOWN. Different combinations of positions represent different signals.

Your task is to read flag signals (two positions) until you receive the STOP signal, then output all the decoded signals on separate lines.

Signal Flag Decoding Table

Left Flag

Right Flag

Signal

UP

UP

DANGER

UP

MID

HELP

UP

DOWN

SAFE

MID

UP

NORTH

MID

MID

STOP

MID

DOWN

SOUTH

DOWN

UP

EAST

DOWN

MID

WEST

DOWN

DOWN

ANCHOR

Input Format

Multiple lines, each containing two strings separated by a space (left flag position, right flag position). Input ends when you decode the signal STOP (which is MID MID).

Output Format

Print each decoded signal on a separate line. Do not print the STOP signal itself.

Example 1

Input:

```
UP UP
UP MID
MID MID
```

Output:

```
DANGER
HELP
```

Explanation:

UP UP = DANGER, UP MID = HELP, then MID MID = STOP (which ends transmission but is not printed).

Example 2

Input:

```
MID UP
DOWN UP
MID DOWN
DOWN MID
MID MID
```

Output:

```
NORTH
EAST
SOUTH
WEST
```

Explanation:

MID UP = NORTH, DOWN UP = EAST, MID DOWN = SOUTH, DOWN MID = WEST, then MID MID = STOP.

Note: When taking input, do not print any message. For example, `input()` or `input("")` would be fine, but `input("Number: ")` or `input("Something")` might fail the tests.

2cooking_catastrophe.md

Problem 17 - Cooking Catastrophe (**)

Charlie is the cook on Henry's latest expedition. He's cooking some dishes for the crew. Unfortunately, he didn't check if the ingredients for each dish were spoiled or not before making the dishes. After realising some of the ingredients were spoiled, he went and reinspected all the ingredients, and collected a list of ingredients that he suspects were spoiled. He knows that serving a dish where any ingredient was spoiled could cause the crew to get food poisoning, thus those dishes can't be served.

Please help Charlie determine the number of dishes that can be served.

Input Format

The first line consists of one integer k - the number of spoiled ingredients.

The second line consists of k strings, s1, s2, s3, …, sn - the names of each spoiled ingredient

The third line consists of one integer n - the number of dishes.

The next n lines consists of one integer ai (the number of ingredients in this dish), followed by ai strings, di1, di2, di3, … - the names of each ingredient in this dish.

Output Format

A single integer, the number of dishes that can be served.

Example 1

Input:

```
1
apple
3
1 apple
2 apple banana
1 banana
```

Output:

```
1
```

Explanation:

The first dish contains apple, which is spoiled.

The second dish contains apple, making it spoiled, despite having other ingredients.

The last dish contains only bananas, which aren't spoiled.

Therefore, only the last dish can be served, so the output is 1.

Example 2

Input:

```
2
apple banana
4
1 banana
2 banana apple
1 mango
2 apple banana
```

Output:

```
1
```

Explanation:

Dishes 1, 2, and 4 are spoiled, as they contain apple and/or banana.

Therefore, only dish 3 can be served so the output is 1.

Example 3

Input:

```
2
kiwi flour
6
3 apple banana chicken
2 banana flour
3 salt kiwi apple
4 tomato bread mayo apple
2 flour yeast
2 chicken egg
```

Output:

```
3
```

Explanation:

Dishes 1, 4, 6 aren't spoiled, so the output is 3.

Note: When taking input, do not print any message. For example, `input()` or `input("")` would be fine, but `input("Number: ")` or `input("Something")` might fail the tests.

2metro_2.md

Problem 7 - Henry's Metro 2 (**)

Henry's Metro system obtained a upgrade. Now, there are 2 lines, which could have intersections.

2 lines on the metro could look like this:

A B C D E F G H I B J K A

Henry starts on a given station and wants to go to an end station. State whether it is possible to go to the end station.

Input Format

The first line contains the name of the start station.

The second line contains the name of the end station.

The third line contains the stations in the first metro line.

The fourth line contains the stations in the second metro line.

Output Format

Print "Y" if it is possible to go to the end station. Otherwise, print "N".

Example 1

Input:

```
A
E
A B C D E F G
H I B J K A
```

Output:

```
Y
```

Explanation:

You can start on the first line and go directly to the 5th station.

Example 2

Input:

```
A
E
A B C
D E F
```

Output:

```
N
```

Explanation:

There is no way to reach the end station.

Note: When taking input, do not print any message. For example, `input()` or `input(“”)` would be fine, but `input(“Number: “)` or `input(“Something“)` might fail the tests.

3cargo_capacity.md

Problem 39 - Cargo Capacity (***)

Captain Kellett needs to load rectangular crates onto his ship's cargo hold floor. The hold has a fixed width and depth, and each crate has its own dimensions when viewed from above. Crates can be rotated 90 degrees if needed.

Before attempting the complex task of arranging the crates, Captain Kellett wants to know if the cargo is potentially packable - that is, whether it's worth attempting to pack them at all, or if it's obviously impossible.

Note: You do NOT need to figure out the actual arrangement of crates. You only need to check if the total area of all crates fits within the hold's floor area, along with other basic feasibility checks.

Your task is to determine if the cargo passes Captain Kellett's feasibility checks. Crates cannot be stacked on top of each other - they must all fit on the same level. Crates can also not be stood on end, their longest edge must make contact with the ground.

Input Format

First line: Two integers hold_width and hold_depth (dimensions of the cargo hold)

Second line: An integer n (number of crates)

Next n lines: Two integers width and depth for each crate

Output Format

Print YES if all crates could potentially fit in the hold, otherwise print NO.

Example 1

Input:

```
10 10
3
2 3
4 2
3 3
```

Output:

```
YES
```

Explanation:

Hold floor is 10 × 10 (area = 100). Crates: 2 × 3 (area=6), 4 × 2 (area=8), 3 × 3 (area=9). Total area = 23. Since total area ≤ hold area, it's possible to fit them.

Example 2

Input:

```
5 5
2
6 3
4 4
```

Output:

```
NO
```

Explanation:

Hold floor is 5 × 5 (area = 25). Crate 1 is 6 × 3 (area=18) - width exceeds hold width even when rotated (3 × 6 still has dimension 6 > 5). Cannot fit.

Note: When taking input, do not print any message. For example, `input()` or `input(“”)` would be fine, but `input(“Number: “)` or `input(“Something“)` might fail the tests.

2kraken.md

Problem 10 - Kraken (**)

Henry Kellett is searching for the Kraken. He receives a coded message from his fellow sailor (your input).

If all the letters K R A K E N - in this order - are found in the message, KRAKEN AHOY! should be printed .

If the letters are not found, WHERE IS KRAKEN? should be printed instead.

(Un)fortunately he did not pay attention to Mr. McMahon’s extensive feedback so he does not care about whether the letters are CAPITAL or lowercase.

Input Format

A single string, the coded message.

Output Format

A single string, either KRAKEN AHOY or WHERE IS KRAKEN?.

Example 1

Input:

```
KELLYROWEDAAWAYKINDLYENJOY
```

Output:

```
KRAKEN AHOY!
```

Explanation:

The 1st, 6th, 11th, 16th, 22nd, 23rd characters of the message form the letters KRAKEN, therefore the all the letters are present, thus the output is KRAKEN AHOY.

Example 2

Input:

```
ARGH IT'S A KRAKE.....
```

Output:

```
WHERE IS KRAKEN?
```

Note: When taking input, do not print any message. For example, `input()` or `input(“”)` would be fine, but `input(“Number: “)` or `input(“Something“)` might fail the tests.

1metro_1.md

Problem 6 - Henry’s Metro 1 (*)

Henry has decided to go on a trip. Instead of taking his ship, he decided to take the metro (you do realise it’s meant to be 1841!? 😂)

A line on the metro could look like this:

Henry Home,John Home,Lok Ma Chau

You are given a line, start station and end station. Including the start and end station, print the number of stations that Henry goes through.

Input Format

The first line consists of a comma separated list of names, representing the stations on the metro line.

The second line consists of a single string, the name of the starting station.

The third line consists of a single string, the name of the ending station.

Output Format

A single integer, the number of stations that Henry goes through.

Example 1

Input:

```
Henry Home,Johns Home,Lok Ma Chau
Henry Home
Lok Ma Chau
```

Output:

```
3
```

Example 2

Input:

```
A,B,C,D,E
B
E
```

Output:

```
4
```

Note: When taking input, do not print any message. For example, `input()` or `input(“”)` would be fine, but `input(“Number: “)` or `input(“Something“)` might fail the tests.

3ship_painting.md

Problem 19 - Ship Painting (***)

Ena is currently drawing Henry’s ship on a canvas. The canvas is a square grid of cells. She draws in a very peculiar way, where she only draws rectangles onto the canvas. She has come up with a list of rectangles she wants to draw onto the canvas, but she wants to know what the final design will look like first.

She tells you that each rectangle she is planning to draw can be specified by 4 numbers. The x-coordinate of the left and right of the rectangle, and the y-coordinate of the top and bottom of the rectangle. (Note that the top left of the canvas has coordinate (0, 0), and the bottom right of the canvas has width (n - 1, n - 1), where n is the width of the canvas.)

Please help Ena determine what her drawing will look like after all the rectangles are painted.

Input Format

The first line consists of two integers n, k - the side length of the canvas, and the number of rectangles she wants to draw.

The next k lines each consist of four integers li, ri, ui, di - the boundaries of each rectangle.

Output Format

The finished design, where . represents unpainted cells, and # represents painted cells.

Example 1

Input:

```
3 2
0 1 0 1
2 2 2 2
```

Output:

```
##.
##.
..#
```

Explanation:

In the first test, after the first rectangle is painted, the canvas looks like this:

```
##.
##.
...
```

Example 2

Input:

```
4 2
0 2 0 1
2 3 1 3
```

Output:

```
###.
####
..##
..##
```

Explanation:

The second rectangle overlaps the first rectangle at coordinate (2, 1).

Note: When taking input, do not print any message. For example, `input()` or `input("")` would be fine, but `input("Number: ")` or `I` might fail the tests.

3port_priority_queue.md

Problem 36 - Port Priority Queue (***)

Captain Kellett's ship has arrived at port and the sailors need to disembark! However, they must leave the ship in order of their rank, with higher ranks leaving first.

Each sailor has a name and a rank number. Lower rank numbers mean higher priority (rank 1 is the highest, rank 2 is second highest, etc.).

Your task is to sort the sailors by their rank and print their names in the order they should disembark.

Input Format

First line: An integer n (the number of sailors)

Next n lines: Each line contains a sailor's name (a string) and their rank (an integer), separated by a space

Output Format

Print the names of the sailors, one per line, in order of their rank (lowest rank number first).

If two sailors have the same rank, maintain their original order from the input.

Example 1

Input:

```
4
Smith 3
Jones 1
Brown 2
Wilson 4
```

Output:

```
Jones
Brown
Smith
Wilson
```

Explanation:

Sorted by rank: Jones (1), Brown (2), Smith (3), Wilson (4)

Example 2

Input:

```
3
Taylor 2
Davis 3
Miller 1
```

Output:

```
Miller
Taylor
Davis
```

Explanation:

Miller has rank 1 (highest priority) so is moved first. Taylor (2) and Davis (3) maintain their original order.

Note: When taking input, do not print any message. For example, `input()` or `input(“”)` would be fine, but `input(“Number: “)` or `input(“Something“)` might fail the tests.

1profit_explosion.md

Problem 32 - Profit Explosion (*)

Henry’s been on a roll plundering different ships recently. He notices that the amount of money he plunders increases by a fixed percentage p every time he plunders a new ship.

Please help henry determine how much total money he makes after plundering n ships.

Input Format

The first line consists of an integer a - the amount of money Henry plunders from the first ship.

The second line consists of an integer n - the amount of ships he plunders.

The third line consists of an integer p - the percent increase between ships.

Output Format

A single decimal number t - representing the total amount of money Henry plunders over all the n ships.

Your answer will be considered correct if the absolute or relative error is less than 10^-6

Example 1

Input:

```
100
3
10
```

Output:

```
331.0
```

Explanation:

The amount of money Henry plunders from the three ships are 100, 110, 121, respectively. Therefore, the output is the sum of these three numbers, 331.

Example 2

Input:

```
15
123
7
```

Output:

```
881236.1044158589
```

Note: When taking input, do not print any message. For example, `input()` or `input(“”)` would be fine, but `input(“Number: “)` or `input(“Something“)` might fail the tests.

2whimsical_names.md

Problem 4 - Whimsical Names (**)

Henry Kellett has a friend called Otto. He realizes that Otto's name is a palindrome! Palindromes are words that look the same when reversed (without counting uppercase or lowercase - for example, oTtO would still be a palindrome.)

Henry wants to know if any other things are palindromes.

Input Format

You are given a single word as the only line of input.

Output Format

A single character, "y" if the word is a palindrome, "n" otherwise.

Example 1

Input:

```
arnold
```

Output:

```
n
```

Example 2

Input:

```
Pop
```

Output:

```
y
```

Note: When taking input, do not print any message. For example, `input()` or `input("")` would be fine, but `input("Number: ")` or `input("Something")` might fail the tests.

1cubic_treasure.md

Problem 26 - Cubic Treasure (*)

After a long journey Henry reaches the mythical city of Cubes. This city is famous for containing cubes of gold that only the best pirates can find. Lucky, Henry is a world class pirate, so he has made his way to the vault of the city. He realises the treasure is all in the form of cubes of diffrent side length, consisting of a precious metal of density d g/cm^3. He wonders if he is strong enough to carry all the treasure out at once.

Please help Henry determine the total mass of all the cubic treasures.

Input Format

The first line consists of two positive integers, n, d - representing the number of cubes and the density of each cube (in g/cm^3) respectively.

The second line consists of n positive integers s1, s2, …, sn, where the i th number represents the side length of the i th cube in cm.

Output Format

A single positive integer - The total mass of all the cubes in grams.

Example 1

Input:

```
3 4
1 2 3
```

Output:

```
144
```

Explanation:

The cubes have volumes of (1 cm)^3 = 1 cm^2, (2 cm)^3 = 8 cm^3, (3 cm)^3 = 27 cm^2.

Thus multiplying by the density of 4 g/cm^3 gives masses of 1 cm^3 x 4 g/cm^3 = 4 g, 8 cm^3 x 4 g/cm^3 = 32 g, 27 cm^3 x 4 g/cm^3 = 108 g, respectively.

Therefore the total mass is 4 g + 32 g + 108 g = 144 g, making the output 144.

Example 2

Input:

```
6 8
4 2 8 11 7 6
```

Output:

```
19792
```

Note: When taking input, do not print any message. For example, `input()` or `input("")` would be fine, but `input("Number: ")` or `input("Something")` might fail the tests.

3scurvy.md

Problem 13 - Scurvy (***)

Henry Kellett's naval fleet is far away when a number of his sailors contract SCURVY. Henry consults Nurse Justin, who tells him that vitamin C is the ultimate cure for scurvy!!!

The crew give him a list of fruit orders that he will need to buy from the nearest island shop. Thankfully, the shop stocks 100 oranges, 100 lemons, 50 grapefruit and 50 pomelos every day.

The shop processes each order one-by-one. Each order reduces the number of a specific fruit the shop has in stock. As soon as the shop cannot fulfil an order, they close for the day. (Note: as his crew are almost all very sick, the shop is likely to close before the orders are finished.)

How many pieces of fruit will be purchased before the shop closes?

Input Format

Input format for each order:

```
name of a fruit
number of fruit needed
```

Output Format

A single integer, representing the total number of pieces of fruit that were sold today.

Example 1

Input:

```
POMELOS
17
ORANGES
50
ORANGES
50
LEMONS
10
POMELOS
35
```

Output:

```
127
```

Explanation:

17 pomelos are purchased (33 remain)

50 oranges are purchased (50 remain)

50 oranges are purchased (0 remain)

10 lemons are purchased (90 remain)

35 pomelos cannot be purchased, so the shop closes.

17 + 50 + 50 + 10 = 127 pieces of fruit were purchased today.

Example 2

Input:

```
LEMONS
1
LEMONS
2
LEMONS
3
ORANGES
2000
POMELOS
39
GRAPEFRUIT
18
```

Output:

```
6
```

Explanation:

1 lemon is purchased (99 remain)

2 lemons are purchased (97 remain)

3 lemons are purchased (94 remain)

2000 oranges cannot be purchased, so the shop closes.

1 + 2 + 3 = 6 pieces of fruit were purchased today.

Note: When taking input, do not print any message. For example, `input()` or `input(“”)` would be fine, but `input(“Number: “)` or `input(“Something“)` might fail the tests.

2sweets.md

Problem 14 - Sweets (**)

Henry is currently at the sweet store trying to buy sweets ready for his next expedition. Each sweet the shop sells costs a certain amount - but alas! Henry cannot buy them all as he only has a certain amount of dollars. Henry is a lover of variety so he will not buy any of the same sweet more than once.

Please help Henry determine the maximum number of sweets he can buy with his money.

Input Format

First line contains two integer inputs: the number of sweets sold by the shop and the amount of money Henry has to begin with.

The third line contains one integer for every sweet sold (the price of each sweet).

[Values are separated by a single space]

Output Format

A single value, the maximum number of sweets Henry can buy.

Example 1

Input:

```
3 2
1 1 100000
```

Output:

```
2
```

Explanation:

Henry can use the 2 dollars to purchase the first two sweets, it can be shown he can't buy any more sweets, thus the output is 2.

Example 2

Input:

```
1 100000
1
```

Output:

```
1
```

Explanation:

The shop only has one sweet, so Henry can buy the only sweet in the shop, thus the output is 1.

Example 3

Input:

```
10 14
5 2 6 3 1 9 3 7 4 2
```

Output:

```
5
```

Note: When taking input, do not print any message. For example, `input()` or `input("")` would be fine, but `input("Number: ")` or `input("Something")` might fail the tests.

2math_quiz.md

Problem 18 - Math Quiz (**)

Dave is currently training to be a quartermaster Henry's ship. The teacher is currently teaching about addition and multiplication. The teacher has given the students a quiz to check their understanding of basic math. Unfortunately Dave wasn't listening to what the teacher was talking about (he was too busy thinking about what's for lunch), which means he doesn't know how to do the quiz.

There are two types of problems on the quiz, addition problems and multiplication problems. Both types consists of a list of numbers, and an operation. The addition problems asks to find the sum total of the list of numbers, and the multiplication problems asks to find the product of the list of numbers.

Please help Dave solve each problem on the quiz.

Input Format

The first line consists of one integer n - the number of problems.

The next n lines consists of one integer ai (the number of numbers in this problem),

followed by one character si which can be either + or * (representing an addition problem or multiplication problem),

followed by ai positive integers (the numbers involved in this problem).

Output Format

n lines, where line i contains the solution to the i th problem.

Example 1

Input:

```
3
3 + 1 2 3
2 * 5 4
4 + 5 5 5 5
```

Output:

```
6
20
20
```

Explanation:

The first problem is an addition problem involving 1, 2, 3. The sum of these numbers is 6, which is the solution to the first problem.

The second problem is a multiplication problem involving 5, 4. The product of these numbers is 20, which is the solution to the second problem.

Example 2

Input:

```
2
1 + 3
1 * 6
```

Output:

```
3
6
```

Explanation:

The first problem adding up the numbers 3, since there is only one number, the answer is just the number itself (3).

The second problem is the same, since the product of a single number is also just itself, the answer is also the number itself (6).

Example 3

Input:

6 4 + 10 15 99 23 3 + 1000 2 7 6 * 2 2 2 3 3 3 7 * 1 2 3 4 5 6 7 2 + 9 99 5 * 1 1 1 1 1

Output:

147 1009 216 5040 108 1

Note: When taking input, do not print any message. For example, `input()` or `input("")` would be fine, but `input("Number: ")` or `input("Something")` might fail the tests.

2chess_at_sea.md

Problem 48 (*) - Chess At Sea

Henry's crew are playing board games at sea. They are being resourceful: as they don't have a chessboard with them, they are using any old bits of marine waste they can find instead of an actual chessboard.

Given two types of sea waste and the dimensions needed, can you help Henry and his crew create the board for their game?

Input Format

First line: A single character - to be used for the white squares on the board.

Second line: A single character - to be used for the black squares on the board.

Third line: The dimensions of the board needed. Note: the board will always be a square.

Output Format

A correctly sized board, made up of alternating squares using characters taken from the input.

Example 1

Input:

```
%
?
3
```

Output:

```
%?%
?%?
%?%
```

Example 2

Input:

🐙
🐟
```
10
```

Output:

🐙🐟🐙🐟🐙🐟🐙🐟🐙🐟
🐟🐙🐟🐙🐟🐙🐟🐙🐟🐙
🐙🐟🐙🐟🐙🐟🐙🐟🐙🐟
🐟🐙🐟🐙🐟🐙🐟🐙🐟🐙
🐙🐟🐙🐟🐙🐟🐙🐟🐙🐟
🐟🐙🐟🐙🐟🐙🐟🐙🐟🐙
🐙🐟🐙🐟🐙🐟🐙🐟🐙🐟
🐟🐙🐟🐙🐟🐙🐟🐙🐟🐙
🐙🐟🐙🐟🐙🐟🐙🐟🐙🐟
🐟🐙🐟🐙🐟🐙🐟🐙🐟🐙

Note: When taking input, do not print any message. For example, input() or input(““) would be fine, but input(“Number: “) or input(“Something“) might fail the tests.

2pi_encryption.md

Problem 9 - Pi Encryption (**)

Henry is trying to bake a pie for his friends birthday. However, to do this he needs help from his friend John. His plan is to send John some coded messages, so he can coordinate the party, but Henry would need a program to encrypt his message. Because John unfortunately did not take IGCSE Computer Science he has no idea on how to program this. Help John encrypt the file with the the following conditions.

Details of how to encrypt the file:

Offset each character by the current digit of π. The first character will be offset by 3, the second by 1, the third by 4 and so on.

Once the digits of π provided (16 of them) run out restart at the front.

The alphabet does not need to wrap around.

“z” offset by 1 is not “a” it’s “{“

Input Format

One string of any length (lower and uppercase).

Output Format

One encoded string, uppercase, lowercase, and symbols.

Example 1

Input:

```
Hello
```

Output:

```
Kfpmt
```

Note: When taking input, do not print any message. For example, `input()` or `input(“”)` would be fine, but `input(“Number: “)` or `input(“Something“)` might fail the tests.

2arctic_pathfinding.md

Problem 33 - Arctic Pathfinding (**)

Captain Kellett is navigating through Arctic waters filled with dangerous icebergs! He needs your help to count how many icebergs are in the area so he can report back to the Admiralty.

You are given a grid representing the Arctic waters where:

. represents open water

X represents an iceberg

S represents the ship's position

Your task is to count the total number of icebergs (X) in the grid.

Input Format

First line: An integer, number of rows

Next rows lines: Each subsequent line contains a string of the following characters

Output Format

Print a single integer: the total number of icebergs in the grid.

Example 1

Input:

```
5
S...X
.X..X
....X
X....
...X.
```

Output:

```
6
```

Explanation:

There are 6 icebergs (X) in the grid.

Example 2

Input:

```
3
S...
....
....
```

Output:

```
0
```

Explanation:

There are no icebergs in this area - it's clear sailing!

Note: When taking input, do not print any message. For example, `input()` or `input(“”)` would be fine, but `input(“Number: “)` or `input(“Something“)` might fail the tests.

3map_markers.md

Problem 37 - Map Markers (***)

Captain Kellett is a master of cryptography and uses clever encoding systems to hide the location of real treasure from pirates. On his navigation charts, he marks many false treasure locations, but only one is real.

Captain Kellett's system uses parity checking: he ensures that every row and column has an even number of treasure markers (T), except for the row and column containing the real treasure, which have an odd number.

Your task is to find the coordinates of the real treasure by identifying which row has an odd count of T markers and which column has an odd count. The intersection of that row and column is where the real treasure is located.

Input Format

First line: Two integers rows and cols (the dimensions of the map)

Next rows lines: Each line contains cols characters representing the map

Map contains: T (treasure marker)

. (empty ocean)

(obstacles)

Output Format

Two integers separated by a space: row col (0-indexed coordinates of the real treasure)

Example 1

Input:

```
4 4
T.T.
....
..T.
T.T.
```

Output:

```
2 2
```

Explanation:

Count treasures in each row:

Row 0: 2 treasures (even)

Row 1: 0 treasures (even)

Row 2: 1 treasure (odd) ✓

Row 3: 2 treasures (even)

Count treasures in each column:

Col 0: 2 treasures (even)

Col 1: 0 treasures (even)

Col 2: 3 treasures (odd) ✓

Col 3: 0 treasures (even)

The row with odd count is row 2, and the column with odd count is column 2. The real treasure is at position (2, 2).

Example 2

Input:

```
3 4
T..T
T...
T..T
```

Output:

```
1 0
```

Explanation:

Count treasures in each row:

Row 0: 2 treasures (even)

Row 1: 1 treasure (odd) ✓

Row 2: 2 treasures (even)

Count treasures in each column:

Col 0: 3 treasures (odd) ✓

Col 1: 0 treasures (even)

Col 2: 0 treasures (even)

Col 3: 2 treasures (even)

The real treasure is at the intersection: row 1, column 0.

Note: When taking input, do not print any message. For example, `input()` or `input("")` would be fine, but `input("Number: ")` or `input("Something")` might fail the tests.

1cruise_ship.md

Problem 23 - Cruise Trip (*)

After a year of hard work, Henry is trying to find the best cruises to relax for a bit. However, being a commander who follows the three ideas of Kindness, Respect and Integrity, he decides to book the cruises for all the sailors.

However, since Henry doesn't want to spend too much money on the trip, he will try to minimise the cost of the trip.

You are given the prices are:

Hong Kong Lines: $150/adult, $100/child, $50/luggage

Gold Ship: $100/adult, $150/child, $125/luggage

International Cruises: $100/adult, $50/child, $150/luggage

Input Format

Three lines of positive integers.

The first line represents the number of adults.

The second line represents the number of children

The third line represents the number of luggage.

Output Format

A single string, depending on which one is cheaper:

Hong Kong Lines is the best option! Gold Ship is the best option!

International Cruises is the best option!

If multiple cruises are cheapest, any one of them would be considered correct.

Example 1

Input:

```
3
2
1
```

Output:

```
International Cruises is the best option!
```

Explanation:

The costs for each cruise are:

Hong Kong Lines: 3 × $150 + 2 × $100 + 1 × $50 = $700

Gold Ship: 3 × $100 + 2 × $150 + 1 × $125 = $725

International Cruises: 3 × $100 + 2 × $50 + 1 × $150 = $550

Therefore the cheapest cruise is International Cruises.

Example 2

Input:

```
1
3
6
```

Output:

```
Hong Kong Lines is the best option!
```

Note: When taking input, do not print any message. For example, `input()` or `input("")` would be fine, but `input("Number: ")` or `input("Something")` might fail the tests.

2wartime_shipping_1.md

Problem 21 - Wartime Shipping 1 (**)

Henry Kellett receives a encrypted message from the admiralty. Before departing, they told Henry Kellett they would be using the Caesar Cipher. Code a device for Henry Kellett to use to decipher the message.

The Caesar Cipher replacing each character in the plaintext with the character that's a certain number of places forward in the alphabet, wrapping around the end of the alphabet if necessary.

Input Format

The first line consists of two characters, representing one possible character of the plaintext, and one character of the corresponding ciphertext.

The next line consists of a single string, representing the ciphertext.

Output Format

A single lowercase string, the plaintext.

Example 1

Input:

```
a h
haahjr avtvyyvd
```

Output:

```
attack tomorrow
```

Explanation:

The first line represents that a was shifted to h during the encryption.

To shift a to h, the encryption shifted all characters 7 places forward. Therefore, if we shift everything backwards by 7 we will get our decrypted text.

Example 2

Input:

```
a x
pefmp xelv
```

Output:

```
ships ahoy
```

Explanation:

The first two characters represents that a was shifted to x during the encryption.

To shift a to x, the encryption shifted all characters 23 places forward. Therefore, if we shift everything backwards by 23 places we will get our decrypted text.

Note: When taking input, do not print any message. For example, `input()` or `input("")` would be fine, but `input("Number: ")` or `input("Something")` might fail the tests.

2cannon_fire.md

Problem 8 - Cannon Fire (**)

Henry Kellett is testing cannon fire at the top of Victoria Peak. He notices that three factors influence the success of his cannonballs: the amount of dynamite used, the strength of the wind, and the amount of rum he has drank today.

When at least 10 sticks of dynamite are used, his cannonballs will not fall into the sea unless one or more of the following facts are true:

today's wind is above 55 knots

Henry has consumed more than 2.5 bottles of rum.

If less than 10 sticks of dynamite are used and there is no wind at all, the cannonballs will:

reach their target if Henry has drank no rum.

explode at Henry's feet if he has drank 4 or more bottles of rum.

In any other circumstance, the cannonball will fall into the sea.

Input Format

Two integers and one float - representing:

a whole number of sticks of dynamite

a whole number of knots (measurement of wind)

a fractional number of bottles of rum drank

Output Format

The message TARGET! should be printed if the cannon fire is successful.

The message SEA. should be printed if the cannonball falls into the sea.

The message FEET! should be printed if the cannonball explodes at Henry's feet.

Example 1

Input:

```
10
57
2.9
```

Output:

```
SEA.
```

Example 2

Input:

```
10
55
1.0
```

Output:

```
TARGET!
```

Note: When taking input, do not print any message. For example, `input()` or `input(“”)` would be fine, but `input(“Number: “)` or `input(“Something“)` might fail the tests.

1transmission.md

Problem 51 - Transmission (*)

Henry’s crew members are missing their family members. Henry, being such a kind captain allows each crew member to send a message back home. He receives 17 messages from his crew members to send back home. However, since sending messages while on a boat is very expensive, Henry would like to know the total length of these 17 messages.

Please help Henry find the total length of these 17 messages.

Input Format

17 lines,

The ith line consists of a single string - the message the ith crew member wants to send.

Output Format

A single integer, the total length of the strings.

Example 0

Input:

```
a short
message
```

.

Output:

```
15
```

Explanation:

The first two lines each contain 7 characters, and the final line contains a single character. The other 14 lines are empty, therefore do not affect the total, which is 15.

Example 1

Input:

This is an example message which does not have very much meaning henry is not holding everyone captive we promise

Output:

97

Explanation:

The lengths of each message are 4, 2, 10, 7, 5, 8, 4, 4, 4, 7, 5, 6, 7, 8, 7, 2, 7 respectively. Thus total length is 97.

Example 2

Input:

Nee onna no ko ni naritai Onegai ii desu ka? Chokoreeto no mori wo kuguru Orenji kabocha no basha ni Noserarete tsurerarete Miruku iro no oshiro Minna kitto akogareteiru Douwa no naka no hiroin Itsuka mita yume ni mita Garasu no hai hiiru Naisho no kimochi honto no kimochi Chotto hanashichaou Wan tsuu surii mahou wo kakete Atarashii boku ni naritai no desu onegai! Yappari boku mo kawaiku naritai na Ano ko mitaku kawaiku naritai na Furiru doresu wo meshimase

Output:

445

Note: When taking input, do not print any message. For example, `input()` or `input("")` would be fine, but `input("Number: ")` or `input("Something")` might fail the tests.

3war_time_shipping_2.md

Problem 22 - Wartime Shipping 2 (***)

As time went on, the admiralty's encoding method got cracked! Desperately, they searched through their archive for a new method and found the Vigenère Cipher! The keyword will change depending on the day of the week but it will always be a five letter word.

Monday: ships

Tuesday: henry

Wednesday: salty

Thursday: bilge

Friday: naval

Saturday: ahoys

Sunday: stern

The first two letters of the encrypted text will be the day.

Decrypt the text!

NOTE: HOW THE Vigenère Cipher WORKS

(contestants have been provided a video)

Input Format

Lowercase string. Punctuations and spaces should not advance key index since they shouldn't be encoded.

Output Format

Lowercase string

Example 1

Input:

```
tuhxgrar mzdguiak
```

Output:

```
h
```

Explanation:

```
Prefix tu shows it is a Tuesday, therefore key used is “henry”
```

Example 2

Input:

```
thxppxi eqo o tbzv sc tptv?
```

Output:

```
where did i park my ship?
```

Explanation:

Prefix th shows it is a Thursday, therefore key used is “bilge”

Note: When taking input, do not print any message. For example, `input()` or `input(“”)` would be fine, but `input(“Number: “)` or `input(“Something“)` might fail the tests.